Bill Splitter
for Roommates!

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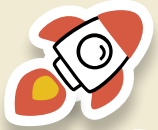

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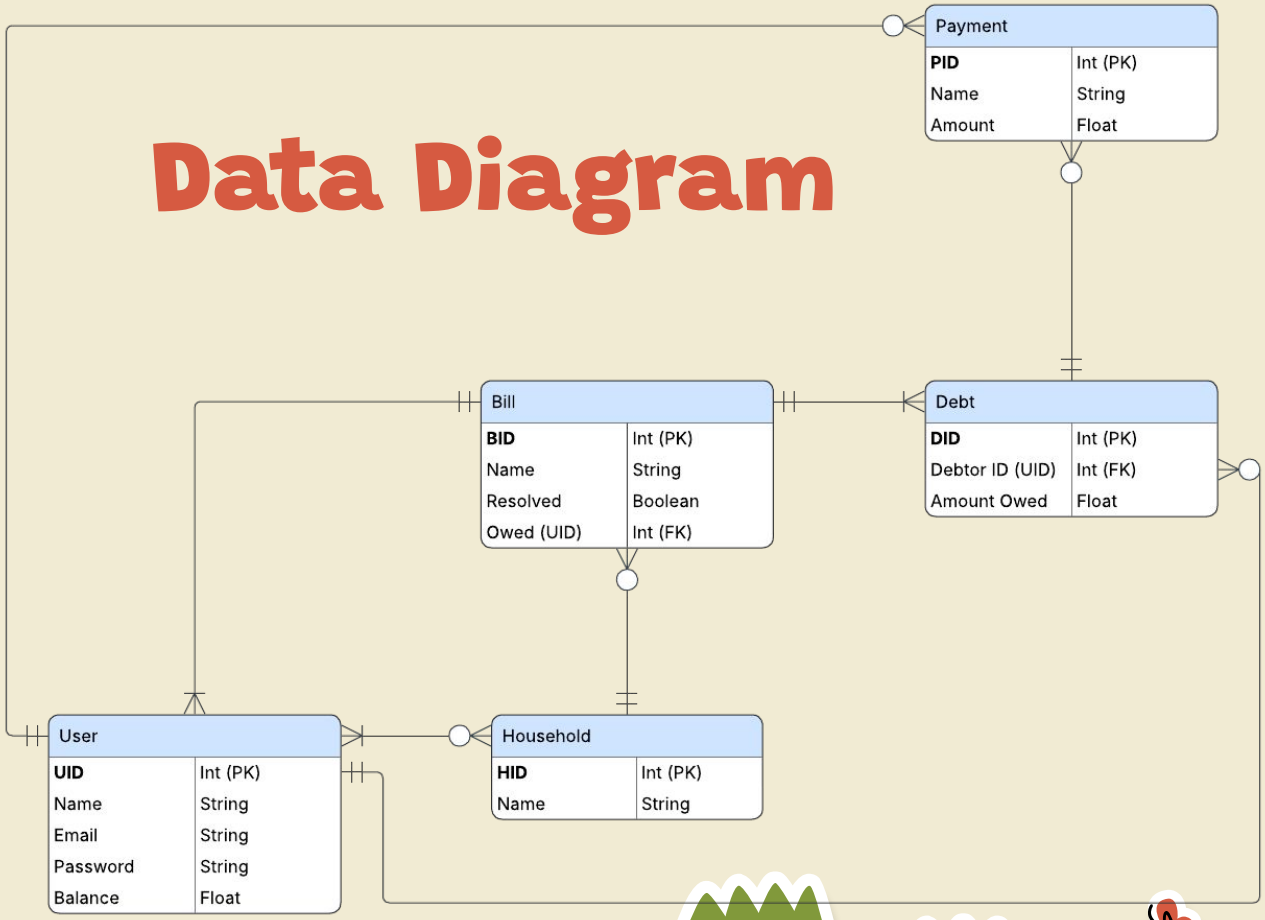
Data Diagram



Frameworks

- **Django** to serve API endpoints and manage DB access
- **Supabase** as the DBMS
- **React** for a dynamic front-end
- **Shadcn** components w/ **Tailwind** for design

End Point Routes

localhost/api/auth/login/ sends login data to backend and gets confirmation

- `login(email: string, password: string)`

localhost/api/households/ retrieve all households

- `getHouseholds()`

localhost/api/bills/ retrieve all active bills and debts that you are involved in

- `getBills()`

End Point Routes

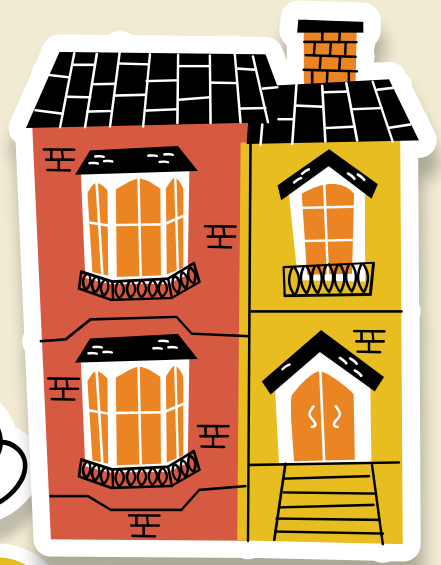
localhost/api/bills-by-person/ retrieve active bills and debts sorted by which household member other than the logged in user they are owed to/by

- `getBillsByPerson()`

localhost/api/bills-resolved/ retrieve all resolved bills and debts that you were involved in

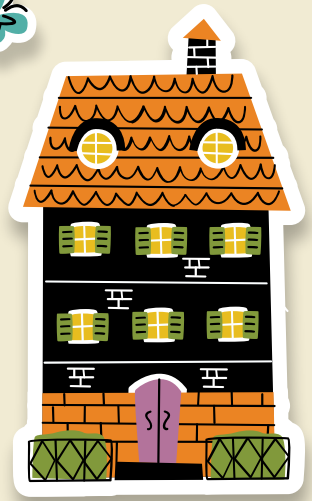
- `getResolved()`

Function signatures



```
login(email: string, password: string):
  Promise<{message: string, token: string}>

getHouseholds(): Promise<Household[]>

getBills(): Promise<{bills: Bill[], debts:
  Debt[]}>

getBillsByPerson(): Promise<[user: User]:
  {bills: Bill[], debts: Debt[]}[]>

getResolved(): Promise<{bills: Bill[],
  debts: Debt[]}>
```

Testing



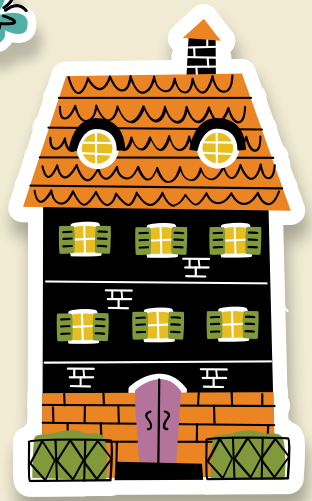
login

Populate user table with the following data:

- **user**: house@gmail.com **password**: 1234
- **user**: hold@gmail.com **password**: 5678

```
login("house@gmail.com", "1234")

login("house@gmail.com", "5678")

login("house@gmail.com", "7777")

login("person@gmail.com", "1234")

login("person@gmail.com", "7777")
```

Testing



getHouseholds

Retrieve all households a user belongs to.
Return empty array if none.

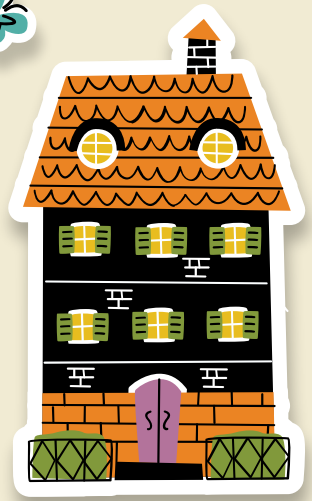
Setup:

Households table:

- H1: “131 Apartment” – members: Alice, Bob
- H2: “Maple House” – members: Alice, Charlie

User table populated with data for Alice, Bob, Charlie, Dave.

Testing



getHouseholds

Logged in user: Alice

`getHouseholds() → [H1, H2]`

Logged in user: Bob

`getHouseholds() → [H1]`

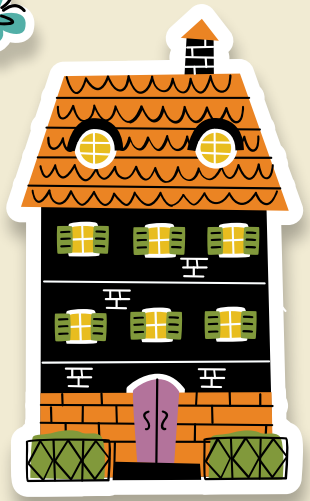
Logged in user: Charlie

`getHouseholds() → [H2]`

Logged in user: Dave (not part of any household)

`getHouseholds() → []`

Testing



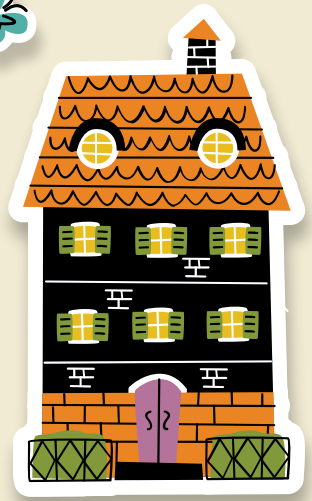
getBills

Household table populated with a new household, User table populated with data for Artichoke, Bart, Chad, and Green

Populate bills/debts tables with the following data:

- **B1** (Bill 1)
  - Creator: Artichoke
  - Debtors: Green
- **B2** (Bill 2)
  - Creator: Bart
  - Debtors: Artichoke, Green

Testing



getBills

Logged in member: Artichoke

`Household.getBills()` → [**B1**], [**B2**]

Logged in member: Bart

`Household.getBills()` → [**B2**], []

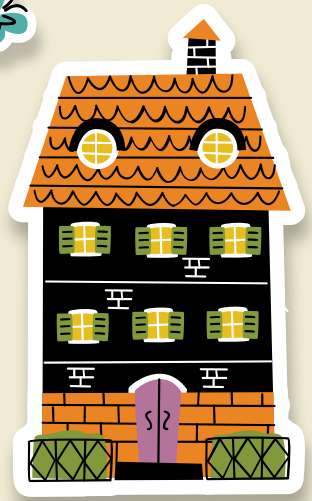
Logged in member: Chad

`Household.getBills()` → [], []

Logged in member: Green

`Household.getBills()` → [], [**B1, B2**]

Testing



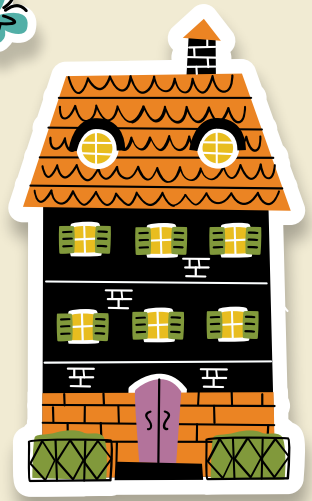
getBillsByPerson

Household table populated with same household, User table populated with data for Artichoke, Bart, Chad, and Green

We populate the Bills/debts tables with the same data:

- **B1** (Bill 1)
  - Creator: Artichoke
  - Debtors: Green
- **B2** (Bill 2)
  - Creator: Bart
  - Debtors: Artichoke, Green

Testing



getBillsByPerson

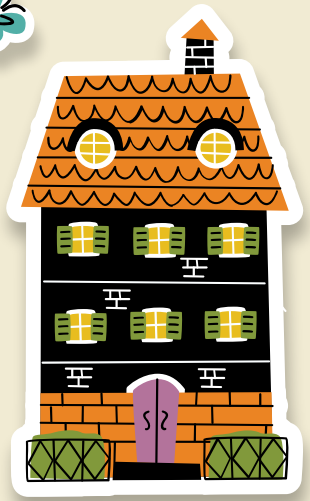
Logged in member: Artichoke

```
Household.getBillsByPerson() →
  [{Bart: ([], [B2])},
   {Chad: ([], [])},
   {Green: ([B1], [])}]
```

Logged in member: Bart

```
Household.getBillsByPerson() →
  [{Artichoke: ([B2], [])},
   {Chad: ([], [])},
   {Green: ([B2], [])}]
```

Testing



getBillsByPerson

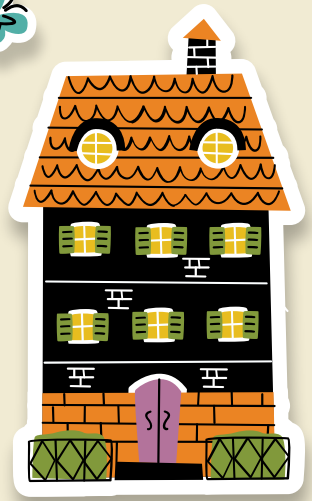
Logged in member: Chad

```
Household.getBillsByPerson() →
  [{Artichoke: ([], [])},
   {Bart: ([], [])},
   {Green: ([], [])}]
```

Logged in member: Green

```
Household.getBillsByPerson() →
  [{Artichoke: ([], [B1])},
   {Bart: ([], [B2])},
   {Chad: ([], [])}]
```

Testing



getResolved

In addition to the existing bills/debts,

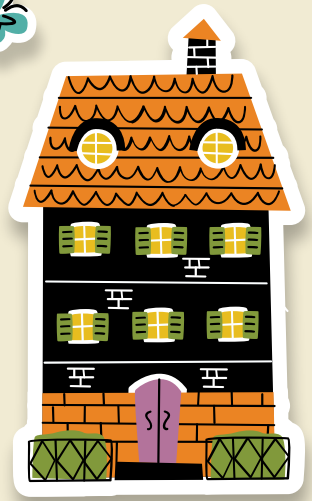
- **B1** (Bill 1)
    - Creator: Artichoke
    - Debtors: Green
- **B2** (Bill 2)
    - Creator: Bart
    - Debtors: Artichoke, Green,

add

- **B3** (Bill 3)
    - Creator: Green
    - Debtors: Bart.

Then, let **B1** and **B3** be resolved.

Testing



getResolved

Logged in member: Artichoke
`Household.getResolved()` → [**B1**], []

Logged in member: Bart
`Household.getResolved()` → [], [**B3**]

Logged in member: Chad
`Household.getResolved()` → [], []

Logged in member: Green
`Household.getResolved()` → [**B3**], [**B1**]

Thank You!